Lab 4 AVCV 2026/2027: Fire-risk classification from satellite images

CUSTOM and FINE-TUNING networks on the FireRisk subset

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1. INTRODUCTION

Given a single RGB satellite image, the task is to predict one of seven FireRisk classes: Water, Very Low, Low, Moderate, High, Very High and Non-burnable. It is an image-classification problem, since the label describes the fire risk of the area rather than the presence of fire. Two properties make it difficult. First, the classes are strongly imbalanced: Water accounts for only 2.4% of the images, whereas Very Low accounts for 31%. Second, the risk levels are ordinal and visually similar, so neighbouring levels such as Low, Moderate and High are easily confused.

We work with a subset of FireRisk [1] containing 5,861 images of 320×320 pixels, split in a stratified way into 3,516 training, 1,172 validation and 1,173 test images whose labels are hidden (Table 3). Both final networks receive 224×224 crops of these images.

We developed two systems. The CUSTOM system is a VGG-style CNN trained from scratch and tuned with a staged random search, while the FINE-TUNING (FT) system adapts ImageNet-pretrained torchvision models, among which Swin-T was selected. Along the way we studied the effect of network width and depth, training length, class weighting, augmentation and image mixing, ensembles, and retraining on train+val.

2. METHODOLOGY

2.1. Preprocessing and data augmentation

Images are read as floating-point values in $[0, 1]$ and normalised with the ImageNet mean and standard deviation. This normalisation is required by the pretrained weights, and we kept it for CUSTOM so that both systems share one pipeline. During training we take a random 224×224 crop of each 320×320 image and then augment every batch on the GPU with horizontal and vertical flips, 90° rotations and random changes of gamma, brightness, contrast and saturation, using either a *light* or a *strong* preset. Validation and test images are only centre-cropped to 224×224, without augmentation or test-time augmentation (TTA).

These augmentations suit satellite imagery: aerial views have no canonical orientation, so flips and rotations create new samples without changing the label, while the colour changes are kept mild so that vegetation cues are preserved. The search selected the *strong* preset for CUSTOM and the *light* one for Swin-T. We also tried mixup [2], CutMix [3], TTA and evaluation at the full 320 px, but none of them gained more than the seed noise of ± 0.008 (Table 2).

2.2. System design: CUSTOM network

The CUSTOM network (Table 5) is built only from primitive layers. It stacks four VGG-style blocks [4], each made of two Conv 3×3–BatchNorm [5]–ReLU layers followed by 2×2 max-pooling, with 48, 96, 192 and 384 channels. Global average pooling (GAP) [6] then feeds a small classifier, FC(384,256)–ReLU–Dropout(0.3)–FC(256,7). The whole network has 2.74 M parameters.

This design differs substantially from the provided reference, which has three blocks of a single 5×5 convolution with 6, 16 and 32 channels, no normalisation and a Flatten head. We kept each change only if it raised validation macro-F1. The first change was to replace the Flatten head with GAP: with Flatten, the first fully connected layer held about 99% of the parameters, and such heads scored lower. We then used two stacked 3×3 convolutions with BatchNorm in each block, which cover the same receptive field as a 5×5 convolution with fewer weights and make training more stable. Finally, widening the network to 48 base channels added +0.03, whereas going further (64 channels, five blocks or larger crops) did not help.

2.3. System design: FINE-TUNING network

Instead of the provided AlexNet [7], we compared three modern backbones of 24–28 M parameters, all with torchvision ImageNet-1k weights [8]: ConvNeXt-T [9], Swin-T [10] and EfficientNetV2-S [11]. EfficientNetV2-S produced a non-finite loss in fp16 and was therefore replaced by ResNet-50 [12]. In every model we replaced the classifier with a new `Linear(in, 7)` layer (768→7 for Swin-T) and the adaptive pooling with a deterministic mean.

No layer was frozen, so backbone and head are trained together. To match the pretraining conditions, the input is 224×224 and normalised with the ImageNet statistics. Because the backbone is already trained while the head starts from scratch, we used two learning rates [13]: a small one for the backbone and, for the head, the same rate multiplied by 1 or 10. Biases and normalisation layers have no weight decay, and stochastic depth [14] adds

further regularisation. Swin-T (swin_p2_1) obtained the best validation macro-F1 (0.546) and was also the most stable run: its training F1 was 0.77, whereas ResNet-50 reached 0.99.

2.4. Training protocol and model selection

Table 1: Training configuration of the selected models.

	CUSTOM (n07_60ep)	FT (Swin-T, swin_p2_1)
Loss	Cross-entropy, class weights $\propto 1/\sqrt{n_c}$, no label smoothing, no mixup	
Optimiser & schedule	AdamW [15]; linear warm-up + cosine decay [16]; fp16 mixed precision	
Learning rate	$3.17 \cdot 10^{-4}$	backbone $2.37 \cdot 10^{-5}$, head $\times 1$
Weight decay	0.0464	0.0395
Batch / warm-up	128 / 1 epoch	32 / 2 epochs
Epochs (selected epoch)	60 (36)	30 (26)
Augmentation	strong	light, drop path 0.2
Common	Seed 42; full train split (no subsampling); Kaggle Tesla T4; PyTorch 2.11 [17]	

Table 1 summarises the training configuration of both selected models. Their hyperparameters come from a random search [18] with median-stopping pruning [19], organised from coarse to fine. For CUSTOM, phases 1–3 ran 24 trials on the architecture and optimiser, 16 trials on secondary settings and three finalists trained for 40 epochs. All of them plateaued at 0.45–0.465 with almost equal training and validation scores, which indicated underfitting. Phases 4–5 therefore tested targeted changes, and only widening the network and training for 60 epochs gave clear gains. A larger 121-trial search reached 0.524 on validation but only 0.47 on the public test, so we discarded it. For FT, we ran 10 configurations per network for 10 epochs and retrained the best two of each network for 30 epochs.

In every run we keep the weights of the epoch with the best validation macro-F1, and the final runs use no early stopping. Training is fully deterministic: we enable `use_deterministic_algorithms` and fix the seeds of all libraries and DataLoader workers, so that training one epoch twice yields bit-identical weights.

3. RESULTS AND ERROR ANALYSIS

Our main metric is macro-F1, the mean of the seven per-class F1 scores, so that rare classes count as much as frequent ones. Every decision was based on validation F1; training F1 served only as a diagnostic, and the public CodaBench score was never used for selection.

Table 2 summarises the experiments. The reference network reached only about 0.36 macro-F1, and our first version with a GAP head reached 0.440. Both, however, scored almost the same on training and validation data, a clear sign of underfitting. The random search confirmed this, as it plateaued at 0.45–0.465. We therefore increased capacity instead, widening the network to 48 base channels with two convolutions per block, and trained for 60 epochs, which produced the final CUSTOM network (n07_60ep) with 0.502. Several other ideas brought no gain:

- inverse-frequency class weights, which made results worse;
- more capacity (64 channels, five blocks) or larger crops;
- CutMix and longer training (80 epochs);
- ensembles, per-class threshold tuning and retraining on train+val.

Changing only the seed already moves results by ±0.008, so smaller differences are not meaningful. For FT, every modern backbone beat AlexNet (0.452), and Swin-T was the best with 0.546.

The learning curves of CUSTOM (Figure 2) show training and validation curves that stay together, with validation F1 saturating at 0.49–0.50 after epoch 30. The model therefore does not overfit; it is limited by the data. Swin-T, in contrast, fits the training set better, with a training F1 of 0.77 against 0.546 on validation. Figure 3 shows the validation confusion matrices (row-normalised %) [TODO: describe once generated].

The per-class results (Table 3, Figure 4) show that Water (0.85) and Non-burnable (0.80) are easy to recognise. The difficult classes are the intermediate levels: Low (0.22), Moderate (0.32) and High (0.33) for CUSTOM, and Low, Moderate and Very High for FT, which points to confusions between neighbouring risk levels [TODO: confirm with Figures 3 and 4]. The difficulty does not seem to come from model capacity: even on its own training images, evaluated without augmentation, CUSTOM reaches only 0.53 (Low 0.30), which suggests that the images themselves carry limited information about these classes. Overall, CUSTOM is small (2.7 M parameters), fast and stable, while FT gains +0.04 on validation at a higher compute cost.

On the public leaderboard (Table 4), our best scores are 0.49 for CUSTOM, obtained with an earlier 40-epoch version of the widened network (p4_09_B), and 0.48 for FT with Swin-T. Every model loses about 0.06 from validation to public, because the same validation images were reused across hundreds of trials and the public subset is small. The best model of the 121-trial search illustrates this: it had the highest validation score (0.524) but only 0.47 on the public subset.

4. CONCLUSIONS

The largest effects came from different sources in the two systems. For CUSTOM, the GAP head, the extra width and the longer schedule together raised validation macro-F1 from 0.44 to 0.50. For FT, the ImageNet pretraining itself was decisive: with a small backbone learning rate, Swin-T reached 0.546 in 30 epochs. Fine-tuning is therefore clearly better on validation, yet both systems score about the same on the public subset (0.49 vs 0.48).

The main limitation lies in the data rather than in the models. The intermediate classes are hard even on training images, and reusing a single validation set for selection inflates its scores by about 0.06. For reference, the original FireRisk paper reports only 61–65% accuracy with about 91k images [1]. Realistic next steps are k-fold cross-validation for a more honest selection, losses that exploit the order of the classes, remote-sensing pretraining and a higher input resolution.

Individual contributions.

- **César Álvarez-Cascos (100475191):** design of CUSTOM search phases 1–3 (two-machine random search, finalists and the 121-trial deep search) and the GPU speed-up of training (images preloaded in RAM, batch augmentation on the GPU, mixed precision).
- **Daniel Truzman (100475161):** design of CUSTOM phases 4–5 (widening to 48×2, 60 epochs, selection of n07_60ep, deterministic Kaggle runs) and the CutMix, ensemble, threshold and train+val experiments.
- **Both:** the training runs on both members' machines, the notebook and the fine-tuning search.

Table 2: Main experiments: validation macro-F1 at the best epoch (L10: mean of the last 10 epochs). Reference and selected models in bold.

Cat.	Change	Objective	Budget	Val F1
CUSTOM	**Reference notebook (StepLR)**	Baseline	[TODO:] ep	≈0.36
CUSTOM	GAP instead of Flatten (first version)	Fewer FC parameters; train F1 ≈0.42	[TODO:] ep	0.440
CUSTOM	Phases 1–3 random search	Architecture, optimiser, regularisation	15 ep (finalists 40)	0.45–0.465
CUSTOM	Inverse-frequency class weights	Help minority classes	[TODO:] ep	L10 0.431 vs 0.453
CUSTOM	48 ch, 2 convs/block (p4_09_B)	More capacity (underfitting)	40 ep	0.496 (L10 0.483)
CUSTOM	64 ch; 5 blocks; crop 288/eval 320	More capacity / resolution	40 ep	no gain
CUSTOM	**60 epochs (n07_60ep)**	Longer schedule	60 ep	**0.502** (L10 0.489)
CUSTOM	CutMix (standard; risk-only)	Regularisation	[TODO:]; 80 ep	L10 0.483; 0.470 vs 0.486
CUSTOM	Deep search (F3_B_038)	Joint search, 121 trials	80 ep, input 320	0.524
FT	**AlexNet (provided recipe)**	Baseline	5 ep	0.452
FT	ConvNeXt-T (cnx_p2_0)	Modern CNN backbone	30 ep	0.505
FT	ResNet-50 (r50_p2_0)	Residual backbone (train F1 0.99)	30 ep	0.539
FT	**Swin-T (swin_p2_1)**	Transformer backbone	30 ep	**0.546**

Table 3: Class counts and per-class validation F1 of the selected models. [TODO: P/R]

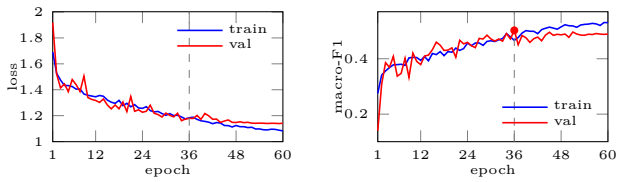
Class	Train	Val	CUSTOM	FT
0 Water	86	29	0.847	[TODO:]
1 Very Low	1,088	363	0.598	
2 Low	535	178	0.221	
3 Moderate	431	144	0.322	
4 High	315	105	0.333	
5 Very High	163	54	0.400	
6 Non-burnable	898	299	0.796	
Total / macro	3,516	1,172	0.502	0.546

Table 4: Public CodaBench macro-F1 (fixed undisclosed test subset). Also: F1_B_037 0.47, swin_p2_0 0.48, r50_p2_0 0.47.

ID	Date	Models (custom / ft)	CUSTOM	FT
1036062	05 Nov	p4_09_B / placeholder	0.49	–
[TODO:]	[TODO:]	n07 (train+val) / [TODO:]	0.494	[TODO:]
[TODO:]	[TODO:]	F3_B_038 / AlexNet	0.47	[TODO:]
[TODO:]	[TODO:]	[TODO:] / swin_p2_1	[TODO:]	0.48
[TODO:]	[TODO:]	[TODO:] / FT ensemble (3)	[TODO:]	0.48

Table 5: Final CUSTOM architecture.

Stage	Layers	Output
Input	RGB crop	3 × 224 × 224
Blocks 1–4	[Conv3×3-BN-ReLU]×2, MaxPool 2×2 (spatial size 112/56/28/14)	48/96/192/384 ch 384 × 14 × 14
GAP	spatial mean	384
Head	FC 384→256, ReLU, Dropout 0.3, FC 256→7	7
2,739,447 parameters; Kaiming initialisation.		



(a) Loss **(b)** Macro-F1

Figure 2: Train (augmented) and validation curves, CUSTOM n07_60ep; dashed: selected epoch 36. [TODO: FT curves]

[TODO: CUSTOM CM]

[TODO: FT CM]

(a) CUSTOM (n07_60ep) **(b)** FT (swin_p2_1)

Figure 3: Validation confusion matrices, row-normalised (%); rows = true, columns = predicted class.

[TODO: one row of misclassified validation images, each with true / predicted label]

Figure 4: Representative validation errors of the selected models (true → predicted class).

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